

Original Report

Patterns of practice for safety-critical processes in radiation oncology in the United States from the AAPM safety profile assessment survey



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Abstract

Purpose: The purpose of this study is to report an overview of the patterns of practice for safety-critical processes in radiation oncology clinics in the United States.

Methods and materials: The Safety Profile Assessment (spa.aapm.org), developed by the American Association of Physicists in Medicine, was released in July 2013. It consists of 92 indicator questions designed to assess the safety and quality of radiation oncology operations. By December 2014, 114 surveys had been completed by clinics within the United States. This database was analyzed to identify those indicators of safety and quality performance with which there was the highest degree of compliance and those indicators with which there was the least. Additionally, we assessed the extent to which key clinical activities were supported by formal policies. Voluntary post assessment surveys were completed by 86 respondents (75%).

Results: The mean number of patients treated per day on external beam radiation therapy devices was 64 (range, 8-600) in the clinics that responded to the survey. The average overall score for the 92 SPA indicator questions was 1.45 (range, 1.00-2.78) on a 5-point scale, with 1 being the most positive. Those indicators that were associated with the highest levels of compliance are dominated by activities that are either strongly recommended, regulated, or associated with revenue generation. Surprisingly, several of those indicators for which there was the least compliance relate to activities that are known to have contributed to serious radiation therapy misadministrations in the past. Formal policies, which are widely regarded as a backbone of a safe clinical system, were

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reported as lacking for some safety-critical procedures.

Conclusions: Although overall this analysis demonstrated reasonable performance across participating departments, several important areas for improvement were identified. The results may guide the allocation of resources both at the level of individual departments and at the professional society level.

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Introduction

The American Association of Physicists in Medicine (AAPM) introduced the Safety Profile Assessment (SPA) in July 2013 (spa.aapm.org). The purpose of the SPA is to provide a practical, resource-efficient, online tool for radiation oncology departments to self-assess their quality and safety profile. The results are benchmarked against a database of participating departments, and each participant can monitor quality or safety improvement projects initiated as a result of the self-assessment. The SPA was developed with significant input from associated professional organizations. A description of the development and pilot testing of the SPA has been published previously.¹ Initial experience has indicated several benefits to the SPA including the value of completing the SPA as a team and its potential ability to foster better communication. The SPA is qualified by the American Board of Radiology as meeting the criteria for practice quality improvement of its Maintenance of Certification Program.²

As of December 31, 2014, there were 132 completed surveys in the database, of which 114 were from departments in the United States. This wealth of information provides a unique opportunity to assess the safety and quality status of radiation oncology in the United States. The identification of common pitfalls and strengths across these sites enables an informed approach for strategy development to mitigate weaknesses, with resources directed to those initiatives with the highest safety/quality return on investment.

Methods and materials

The SPA has been fully described elsewhere.¹ Briefly, it is an online tool for a radiation oncology department to self-assess against a suite of 92 performance indicators. The indicators cover the following aspects of a department's performance: institutional culture, quality management, managing change and innovation and clinical performance. The section on clinical performance indicators, which is the largest of the 4 sections with 51 questions, also queries whether there is a written procedure document available for the process in question. The survey leader enters their assessment of the degree of compliance with a particular performance indicator using a 5-point Likert scale ranging from "always/strongly agree" to

"never/strongly disagree," together with an option to enter "not applicable." The information in the database is fully anonymous, to encourage accurate reporting. Neither the developers nor the database managers at AAPM headquarters can associate a survey response with an institution or with any particular user.

The analysis presented here is based on response data from 114 US institutions over the period from July 2013 to December 2014. The Likert scale was converted to a numerical scale from 1 to 5, with 1 corresponding to "always/strongly agree." Where there was no answer or the response was "not applicable," those data were not included in the analysis. The data do include some incomplete responses, that is, surveys in which the respondent did not provide answers for all of the 92 questions in the SPA. Twenty-two respondents (19% of the total) omitted answers to 10 or more questions. This is in part a reflection of practice, since not all questions were relevant to all practices. When questions specifically related to brachytherapy or stereotactic body radiation therapy are excluded, only 14 respondents (12%) omitted 10 or more answers. There was no correlation between mean score or variability with number of omitted answers.

At the completion of the SPA, participants are asked to fill out a survey that includes questions about their clinic and their experience with the SPA. This provided information on the practice environment of respondents and feedback on the utility of the tool.

We have found that the data are well described by a Poisson distribution, and these were fit via a routine in MATLAB that returns the Poisson λ parameter, which is also equal to the mean. From this, it follows that the variance of each response data set is equal to the mean. A 2-tailed Mann-Whitney test was used to assess score difference with a significance of .05. In the relevant tables that follow, we present mean scores consistent with the Poisson distributions observed to fit the data. Other data, including standard deviations, can be found in the supplementary materials accompanying this article (available online only at www.practicalradonc.org). Readers who are interested in undertaking further analysis of this rich database can access the data there.

Results

Table 1 shows the numbers of indicator questions in each section of the SPA together with the mean score by

Table 1 Sections of the SPA tool and mean scores within each

Section	Name	No. of indicators	Mean score
I	Institutional Culture	18	1.99
II	Quality Management	15	1.69
III	Managing Change and Innovation	8	1.90
IV	Clinical Performance Indicators	51	1.54

"Never/strongly disagree" corresponds to a score of 5, whereas "always/strongly agree" is a score of 1.

SPA, Safety Profile Assessment.

section. At the $P < .05$ level, the only statistically significant difference was between sections I and IV. Figure 1 shows the distribution of scores for the 114 completed surveys analyzed here. The peak of the distribution lies between performance levels "always/strongly agree" and "most of the time/agree." The score for all questions averaged across all clinics was 1.45 (range, 1.00-2.78).

The key results from the SPA database of responses from US clinics are presented in Tables 2 through 5. Tables 2 and 3 show those performance indicators with the greatest compliance (lowest scores) and those with the least compliance (highest scores), respectively. The questions with the lowest average score also tended to show the least variability in their scores, which is also a property of the Poisson distribution that appears to describe these data well.

When we examined the 51 clinical performance indicators (section IV of the SPA), several patterns emerged. First, there appear to be differences between the indicators for brachytherapy versus those for external beam radiation therapy. The performance indicators for brachytherapy yielded responses of "always/strongly

agree" (score 1) far more often than the performance indicator categories for external beam radiation therapy. Fifty respondents answered with the most favorable score of 1 for the brachytherapy questions, whereas among the 5 external beam radiation therapy categories, the highest number of respondents with a score of 1 was 42 (for the question about posttreatment completion).

In the clinical performance indicators section, respondents are also asked to indicate a yes/no response to the following: "Our department has a formal policy for this." A response is requested for each of the 51 indicator questions. The data are shown for the most frequent positive responses (Tables 4) and the least frequent positive responses (Table 5). Not all SPA surveys included responses to this formal policy statement. On average, 87 (76% [range, 66%-87%]) of those taking the survey answered the questions about policy. Both the numbers of positive (yes) and negative (no) responses are shown in Tables 4 and 5. In Table 4, an average of 87% of those responding answered with a definitive positive response, whereas according to the data in Table 5, an average of 53% of those responding to these questions answered with a definitive negative response.

Of the 114 respondents, 86 (75%) completed the voluntary post-SPA survey, which provides basic information about the clinical environment and their experience with the SPA. In these responses, the mean number of patients treated per day on external beam radiation therapy devices was 64 (range, 8-600). The majority of participants (53%) completed the SPA as a team, with an average group size of 5.9 people (range, 1-78). Sixty-six percent rated the SPA as easy or very easy to complete, and 67% indicated that they intended to complete the SPA again in the future. These data are consistent with observations during pilot testing of the SPA.¹

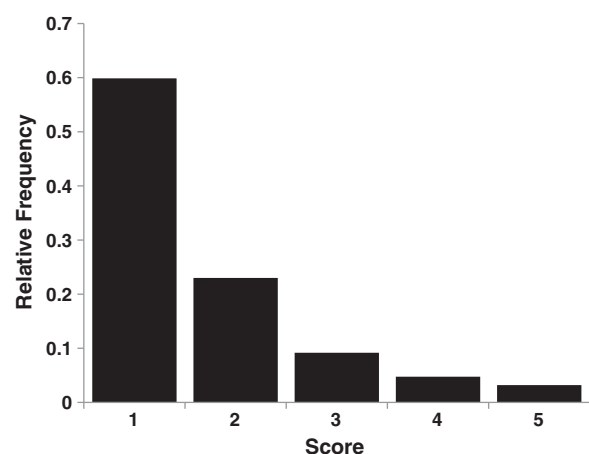


Figure 1 Distribution of scores for the 114 completed Safety Profile Assessment responses. "Never/strongly disagree" is represented by a score of 5 and "always/strongly agree" by a score of 1.

Discussion

The data presented here provide a snapshot of safety-critical practices in radiation oncology clinics in the United States from the first year of experience with the SPA tool. Several interesting patterns emerged. First, of the 4 main sections of the SPA, the only significant difference in scores was between section I (institutional culture) and section IV (clinical performance indicators), with the more well-established clinical performance indicators scoring more positively than the questions concerning institutional culture. This may be attributable to a variety of factors, including the fact that safety culture is an emerging concept for many providers. To achieve advances in safety and quality, a positive safety culture must be cultivated in equal measure with the more established standards of clinical practice. As experience matures, it will be important to determine whether sections

Table 2 The 10 performance indicators with which there was the highest level of compliance (lowest scores)

Performance indicator	Mean score	Section
31. Dosimetry equipment is calibrated every 2 years by an accredited dosimetry calibration laboratory.	1.03	II
64b. Pretreatment patient-specific dose verification is performed for the following treatment modalities: b. IMRT QA.	1.06	IV
85. The Authorized User approves the plan and written directive before treatment. (Brachytherapy)	1.11	IV
75. Staff maintains visual and audio contact with patients throughout treatment.	1.12	IV
26a. Pre-clinical validations are performed for: a. Treatment delivery systems.	1.13	II
83. The source strength is verified prior to clinical use. (Brachytherapy)	1.14	IV
86. The location of the source(s) is verified immediately after treatment. (Brachytherapy)	1.14	IV
63. An initial physics plan review is completed consistent with the appropriate guidelines.	1.17	IV
77. Physics chart checks are performed weekly.	1.18	IV
51. Patient identity is verified before simulation.	1.19	IV

“Never/strongly disagree” corresponds to a score of 5, whereas “always/strongly agree” is a score of 1.

IMRT, intensity modulated radiation therapy; QA, quality assurance.

I and IV of the SPA continue to demonstrate significance differences, and if so, what can be done to address it.

A second pattern concerned performance indicators. For the indicators with the greatest compliance, 2 characteristics were dominant (Table 2). The highest-ranked compliance levels were associated with items that are either regulated (performance indicators 31, 83, 85, and 86, with the latter 3 referring to brachytherapy), billable (performance indicators 64, 63, and 77), or considered good practice as established in recommendations from the American Society for Radiation Oncology (ASTRO)/American College of Radiology and elsewhere. The association of these high-compliance indicators with regulations, recommendations, and reimbursement is perhaps not surprising. However, even for these positive-scoring indicators, compliance was not universal across participating centers.

In contrast, those indicators for which there was the least compliance did hold some surprises. There was more variability between centers with these more negative-scoring indicators. In general, there were fewer well-established recommendations around these indicators, which perhaps explains the higher (more negative) scores and greater

variability. Some of these indicators have well-established recommendations, however, which raises some concerns. These include physician peer review of plans before treatment (performance indicator 65) (Table 3), for which there was poor compliance. This is in spite of strong recommendations by ASTRO in the form of safety white papers³ and the “Safety Is No Accident” report from ASTRO.⁴ Also lacking in compliance was the collection and analysis of near-miss incident data (performance indicators 13 and 15) (Table 3). This has also been emphasized recently by professional bodies and the literature on safety in radiation therapy, including via specific recommendations by ASTRO.⁵ Modern incident learning systems, such as RO-ILS: Radiation Oncology Incident Learning System and the IAEA’s Safety in Radiation Oncology (SAFRON) system,⁶ are actively soliciting the reporting of near-miss events. Incident learning is recognized as being invaluable in proactively making radiation therapy safer.

A final area of poor compliance was in the risk assessment of new clinical systems (performance indicator 39) (Table 3). Although there are no formal recommendations concerning this, the poor compliance may reflect

Table 3 The 10 performance indicators with which there was the lowest level of compliance (highest scores)

Performance indicator	Mean score	Section
52. A time out is performed at simulation.	2.23	IV
20. Standard operating procedures for safety-critical clinical processes are reviewed regularly.	2.25	II
79. Therapists perform weekly chart checks.	2.27	IV
24. Clinical staff competencies are reviewed regularly.	2.29	II
38. An independent review of commissioning results is performed prior to implementation of new clinical systems and processes.	2.32	III
90. A therapist performs a final chart check.	2.33	IV
39. Potential risks associated with the introduction of new clinical systems and processes are assessed prior to implementation.	2.34	II
65. Physician peer review of new treatment plans occurs within the first week of treatment.	2.43	IV
15. The radiation oncology department formally reviews reports of near-misses.	2.59	I
13. Clinical staff submits written reports of near-miss incidents.	2.78	I

“Never/strongly disagree” corresponds to a score of 5, whereas “always/strongly agree” is a score of 1.

Table 4 The clinical performance indicators for which a positive answer was most frequently given to “Our department has a formal policy for this”

Clinical performance indicator	Yes	No
68. Patient identification is verified prior to each treatment.	91	8
77. Physics chart checks are performed weekly.	87	8
63. An initial physics plan review is completed consistent with the appropriate guidelines.	82	13
64. Pretreatment patient-specific dose verification performed.	81	14
51. Patient identity is verified before simulation.	80	14
75. Staff maintains visual and audio contact with patients throughout treatment.	80	12
70. The staff acquires portal imaging and/or isocenter images in accordance with published guidelines.	80	12
91. A physicist performs a final chart check.	78	14
76. Physicians routinely review localization images.	78	14
78. Physicians perform weekly treatment management visits.	77	17

The total number of “yes” and “no” answers are shown. Not all of the 114 respondents answered each question.

the newness of approaches such as Failure Modes and Effects Analysis in the radiation therapy community. There are a number of studies now reported that describe the use of Failure Modes and Effects Analysis in a clinical radiation therapy context⁷⁻⁹ and in other areas of healthcare.¹¹ Active engagement of radiation therapy practitioners⁸⁻⁹ in risk-based quality management will be required if this observed weakness in current practice is to be addressed, especially in light of the forthcoming report of the AAPM’s Task Group 100.¹⁰ The poor compliance in these indicators may be the result of a lag between the formulation of recommendations and their wider adoption in the community.

Independent verification of commissioning data (indicator 38) also assumed a low priority in many of the institutions participating in the SPA tool. This is sobering given that errors at commissioning have been single points of failure in several catastrophic radiation therapy misadministrations. It may be that for smaller centers, independent checking of commissioning data is less practical than in those institutions with many physicists. Vendors and professional organizations need to consider easily accessible repositories of accurate data that can be used to validate locally generated commissioning results.

An interesting apparent contradiction exists between the results presented in Table 2 and those in Table 3. Although “Patient identity is verified before simulation” is used frequently, “Time out is performed at simulation,” is performed relatively infrequently. It is possible that responders encountered interpretation difficulties in distinguishing between these 2 statements, although it is also possible that it is generally only patient identification that happens before simulation commences and not more structured confirmation of the scan volume and other clinically relevant information.

Standard operating procedures are another cornerstone of safety in high-reliability industries,¹¹ and clearly, to remain relevant, these need to be reviewed on a regular basis; however, this important function appears to not be well established in many centers. Tables 4 and 5 refer specifically to the extent to which clinical processes are documented (ie, a standard operating procedure has been developed). The statement in the survey tool is, “Our department has a formal policy for this,” and yes or no answers are the only 2 options. From Table 5, it can be seen that more than half of the institutions that responded do not have formal policies for several clinical processes

Table 5 The clinical performance indicators for which a positive answer was least frequently given to “Our department has a formal policy for this”

Clinical performance indicator	Yes	No
56. Import of complementary imaging for planning includes verification of patient orientation.	32	52
47. Curative intent cases undergo multidisciplinary review to determine treatment options.	32	50
55. Site and side are verified with a secondary source document at the time of planning.	36	50
73. The physician treatment directive specifies motion management strategies to be used where appropriate.	37	41
60. The impact of previous radiation treatments on the current treatment plan is evaluated by both the planner and the physician.	39	48
57. Electronic transfer of patient information from simulation to planning system is verified for each patient.	39	47
59. The physician communicates patient-specific planning goals to the treatment planning team.	40	47
80. Prescription revisions are communicated to the involved team members at time of revision.	44	44
58. Patient information is verified for all data used for treatment planning.	45	43
49. The staff adheres to a guideline for managing IV contrast reactions.	46	29

The total number of “yes” and “no” answers are shown. Not all of the 114 respondents answered each question.

that are known to be hazardous. Even the indicators that more frequently have a policy in place do so only two-thirds to three-quarters of the time. For example, standard operating procedures to guard against laterality errors, which are far from unknown,¹² are lacking in most centers. Also of note, institutions left the policy component of indicators blank an average of 24% of the time, which suggests that the number of actual policies in place is lower than we report here. Poor documentation and adherence to policy are problematic. Not only do respected patient safety organizations¹¹ recognize that standard operating procedures are highly effective means for reducing the risk of untoward clinical events, but there is also evidence of this from incident learning systems.¹³

Given that policies and procedures appear to be a weak point for some organizations, there may be an opportunity for professional organizations to assist. It may be possible to develop boilerplate policies and procedures documents that could then be customized to an individual center's particular practice. Of course, having an appropriate standard operating procedure is only part of the issue; the other part is ensuring that staff actually follows the procedure fully. There is a much better chance of procedures being followed if they are written clearly, their relevance to safety and quality is explicit, and they are endorsed and promoted by management.

Figure 1 presents the aggregate safety profile across institutions participating in the SPA database. Although our foregoing discussion has identified weaknesses, some of them potentially serious in current practice, the overall picture presented in Fig 1 is encouraging. The distribution of centers' average scores is centered toward the "always/strongly agree" end of the Likert range. A future direction is to reduce the relatively large spread in the scores and bring more centers into compliance.

This study has limitations that need to be appreciated in reviewing the results and discussion presented here. Contributing to the SPA database is purely voluntary, so the distribution across types of center (large vs small, academic vs nonacademic, etc) may not be fully representative of the practice of radiation oncology in the United States. The post-SPA surveys, however, which were completed by 75% of respondents here, show that the mean number of patients treated per day on external beam radiation therapy devices was 64. This indicates that the average respondent here is from a medium-sized clinic, likely with 2 treatment units, which is reflective of much of the practice nationally.

However, it is acknowledged that the data presented here may be biased toward those centers that place more emphasis on quality and safety. If this is so, the actual practice nationwide may be less positive than the results presented here.

Another limitation of this study is the possibility that data presented here may include multiple responses from an institution. Although it is not possible to assess or control for this, given the anonymous structure of the SPA,

we expect the effect is small, because a majority of centers are completing the SPA as a group.¹ In this context, it is unlikely that more than 1 person from a center would complete the SPA. The data presented here also include some incomplete surveys, that is, surveys in which the respondent did not answer all of the 92 SPA questions. This does not appear to impact the conclusion, however, because it is related in part to specialty practices, and there was no observed relation between the number of omitted answers and scores.

A final point to note about this study is that there are 92 indicator questions in the SPA, and they cannot all be equally important for quality and safety in the clinic. We have presented the results as they are in the database and have attempted to avoid any implicit prioritization among the 92 indicators in the discussion. We leave it to the reader to form his or her own opinion as to the relative importance of the various indicators.

Conclusions

The AAPM's SPA provides a unique opportunity to assess the safety and quality landscape of radiation oncology in the United States. Of the 114 centers participating thus far, quality/safety profiles are toward the positive end of SPA's Likert scale, albeit with considerable variation. There is good compliance with well-established safety recommendations but poorer compliance with recommendations that are more recent and emerging. Formal policies for some important indicators are lacking. As further data are collected, more detailed analysis of the rich data set will allow the identification of systemic weaknesses and suggest, where possible, strategies for addressing these at the national and local levels.

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